IMAGE PROCESSING · LOCAL OPERATIONS

Local histogram processing

How neighborhoods, filters, and derivatives turn raw pixels into clearer images.

THE LOCAL VIEW

From one pixel to its neighborhood

A single intensity can be ambiguous. Nearby pixels provide the context needed to decide whether it belongs to a shadow, a texture, or an edge.

Point processing

Changes one pixel using only its own intensity.

Local processing

Uses a neighborhood around each pixel to reveal context.

Moving window

The neighborhood is usually a small rectangular window that moves across the image.

LOCAL EVIDENCE

A local histogram describes what surrounds a pixel

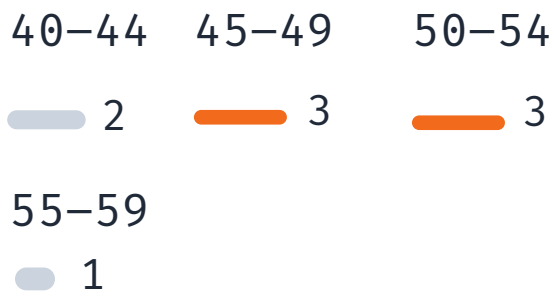
For each window, count intensity values, inspect the distribution, and transform the center pixel using that local evidence.

ACTIVE WINDOW

42	45	47
44	51	53
48	50	55

The center pixel is interpreted with its eight neighbors.

LOCAL HISTOGRAM



The distribution tells us where this neighborhood's intensity mass sits.

Spatial filtering is a weighted neighborhood operation

A mask assigns weights to nearby pixels; the window moves across the image and computes one new output value at each location.

LINEAR

Depend on sum of product

blurring , edge detection and sharpening.

NON LINEAR

Depend on sorting algorithm

min , max, median and midpoint.

PIXELS

2	4	6
3	5	7
1	4	8

×

MASK

0	1	0
1	4	1
0	1	0

Each position produces one output value from the local products and their sum.

ORIENTATION MATTERS

Correlation and convolution differ by one rotation

Both operations slide a mask and compute the same sum of products. Their distinction is the mask's orientation before the slide begins.

CORRELATION

Use the mask as written

a	b	c
d	e	f
g	h	i

Slides the mask as written and sums products.



180°

CONVOLUTION

Rotate the mask first

i	h	g
f	e	d
c	b	a

Rotates the mask by 180° before the same sum-of-products operation.

For symmetric masks, the two results are identical. For asymmetric masks, orientation matters.

THE CORE OPERATION

Correlation is a sliding neighborhood operation

A small kernel moves across the image. At each location, multiply the overlapping values, then add them. The output summarizes one local pattern.

01

Place

Center the kernel on one pixel.

02

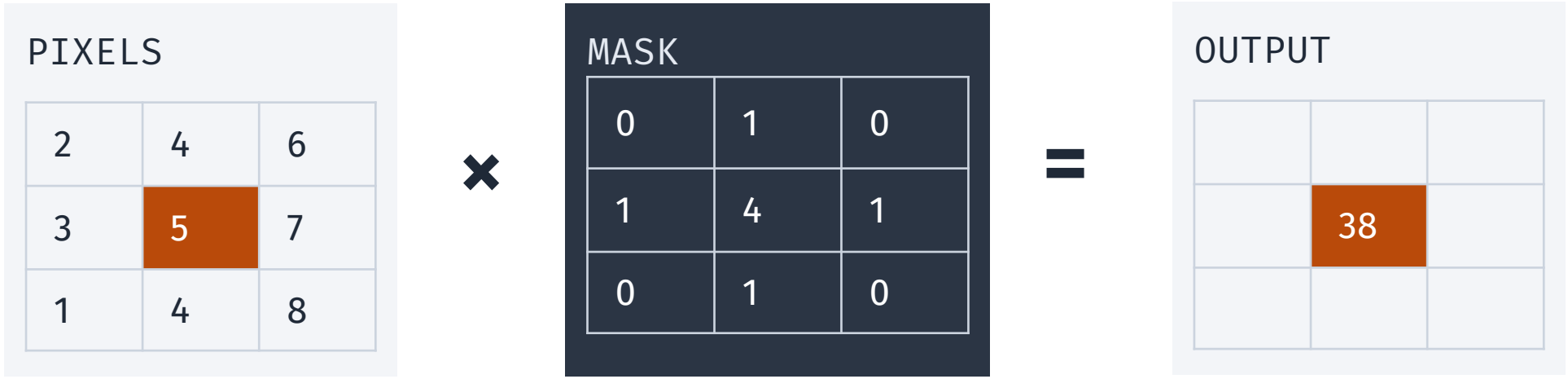
Multiply

Pair each kernel value with its neighbor.

03

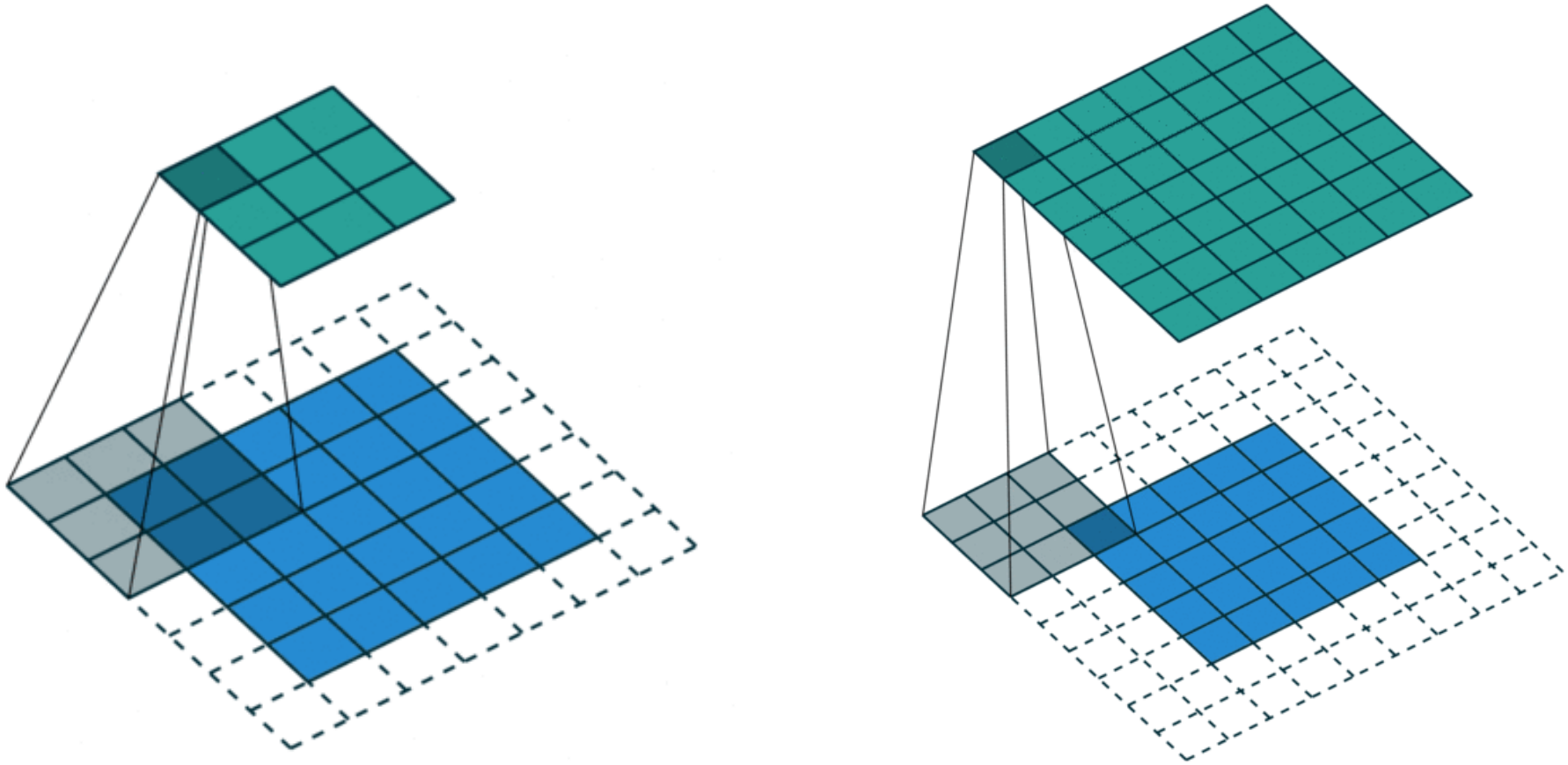
Sum

Write one output value.



THE CORE OPERATION

Correlation is a sliding neighborhood operation



THE FIRST FAILURE

The boundary problem: the neighborhood runs off the image

Interior pixels have a complete neighborhood. Border pixels do not. The filter needs values outside the image, but those values do not exist yet.

INTERIOR

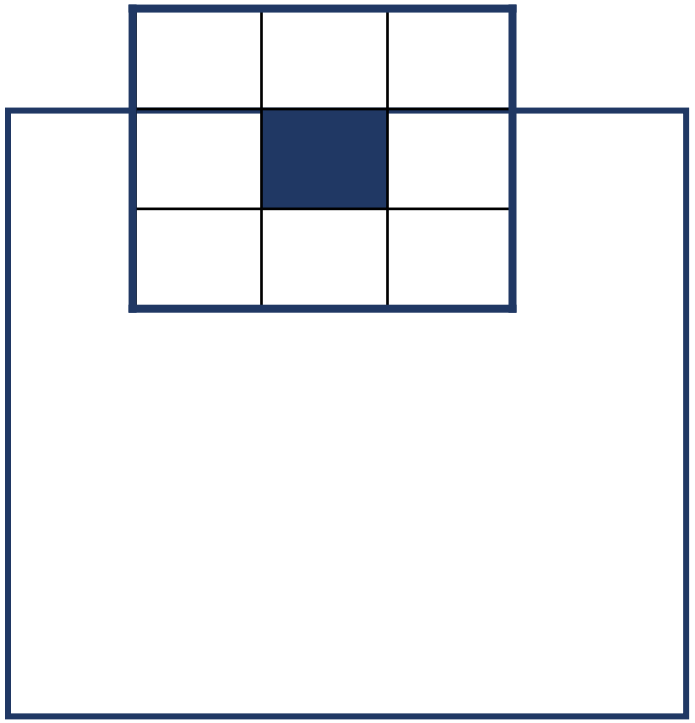
Every tap lands on a real pixel.

BORDER

Some taps land outside.

RESULT

Without a rule, the output may change size.



FOUR VALID RULES

Padding supplies the missing context

Padding extends the image before correlation runs. The choice changes what the filter “sees” at the border.

ZERO

Add black outside.

Simple, but it can create dark border responses.

REPLICATE

Repeat the nearest edge.

Edge values continue outward.

IGNORE

Ignore the edges border.

result an image smaller than the origional image.

Padding supplies the missing context

5	82	90	150
120	30	70	80
10	60	50	20
40	140	11	40

Original image

0	0	0	0	0	0
0	5	82	90	150	0
0	120	30	70	80	0
0	10	60	50	20	0
0	40	140	11	40	0
0	0	0	0	0	0

After zero padding

5	5	82	90	150	150
5	5	82	90	150	150
120	120	30	70	80	180
10	10	60	50	20	20
40	40	140	11	40	40
40	40	140	11	40	40

After duplicate/ replicate

30	70
60	50

Ignore border

LINEAR FILTERS

Linear filtering combines pixels with weighted sums

$$G(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)\, f(x+s, y+t)$$

The output is a weighted sum of neighborhood pixels. The weights define what the filter preserves, suppresses, or emphasizes.

Average

Box filter

Equal weights spread influence across the window.

SMOOTH

MASK

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

Average

Weighted filter

Nearby pixels receive more weight than distant ones.

SMOOTH

MASK

0	$\frac{1}{8}$	0
$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$
0	$\frac{1}{8}$	0

Linearity makes these filters predictable and easy to analyze in frequency terms.

THE SMOOTHING TRADE-OFF

Averaging reduces noise, but softens detail

A mean filter replaces the center pixel with the neighborhood average. Larger windows smooth more aggressively, but they also mix information across boundaries.

01

01 · INPUT

Noisy patch

Random intensity changes interrupt an otherwise coherent region.

02

02 · FILTER

Neighborhood average

The center value is replaced by the mean of nearby pixels.

03

03 · RESULT

Smoother, less sharp

Noise falls, but edges blur because pixels from both sides of the boundary are mixed.

LINEAR FILTER

Average filter example

PIXELS

10	10	10
10	200	10
10	10	10

×

MASK

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

=

OUTPUT

	30	

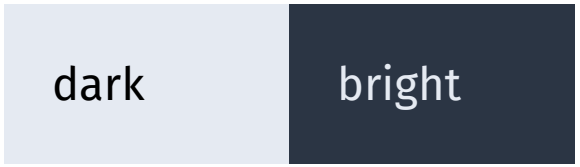
(8/9)*10+(200/9)=30

EDGE DETECTION

Edges are places where intensity changes quickly

An edge is a location where brightness changes quickly. Gradient filters measure that change in all directions.

STEP EDGE



Intensity changes abruptly at the boundary.

Sobel mask

-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

2	1	0
1	0	-1
0	-1	-2

0	-1	-2
1	0	-1
2	1	0

Robert mask

0	1	0
0	0	0
0	-1	0

0	0	0
1	0	-1
0	0	0

1	0	0
0	0	0
0	0	-1

0	0	1
0	0	0
-1	0	0

Sharpening filter

positive and negative.

Sum =1

Post process

- Absolute
- Clipping (<0----------0 & >255----------255)

Laplacian filter

0	1	0
1	-4	1
0	1	0

Laplacian filter

1	1	1
1	-8	1
1	1	1

Sharpening filters

0	1	0
0	1	0
0	-1	0

0	0	0
1	1	-1
0	0	0

0	0	1
0	1	0
-1	0	0

1	0	0
0	1	0
0	0	-1

Example

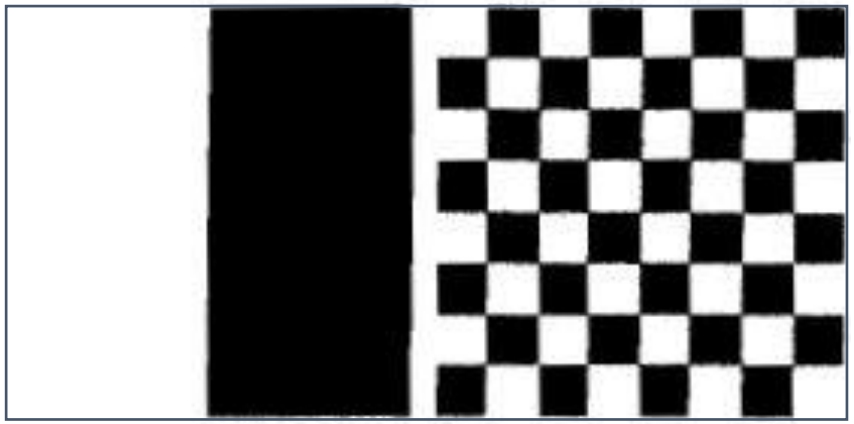
1) Laplacian with -8 in the center yields sharper result than one with a -4 in the center
   explain reason in detail

2) Suppose that each image is blurred with a 3*3 averaging mask

   a) The histogram of the following images are the same?

   b) Would the histograms of blurred images equal?

   a& b → explain and sketch



Example cont.

3) Using the figure to the right, find the filtered image using zero padding of the original image

2	9	10	0
7	1	6	1
10	15	2	6
11	3	8	10

Original image

1/4

0	1	0
1	0	1
0	1	0

Shift – invariant
Linear filter

?	?	?	?
?	?	?	?
?	?	?	?
?	?	?	?

Filtered image

Example cont.

4) Predict the output using the following correlation filters for the image in the following figure?

